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Math 215A

Homework 5
Due: October 28, 2026

Rings are understood to be commutative, unless stated otherwise.

Problem 1

Give an example of a ring A and a multiplicatively closed subset $S \subset A$ such that the relation "$(a, s) \sim (b, t)$ if $at - bs = 0 \in A$" on $A \times S$ is not an equivalence relation. [This explains the more complicated definition of $A[S^{-1}]$, in general. Make sure you know that definition.]

Solution. Consider $A = \mathbb{Z}_6$ and $S = \{1, 3\}$. Consider $(0, 3), (0, 1), (2, 3)$.

$$(0) * 1 - 3 * 0 = 0 \Rightarrow (0, 3) \sim (0, 1)$$
$$0 * 3 - 3 * 2 = 6 = 0 \Rightarrow (0, 3) \sim (2, 3)$$
$$0 * 3 - 1 * 2 = -2 = 4 \neq 0 \Rightarrow (0, 1) \text{ is not equivalent to } (2, 3).$$

Thus $\sim$ is not transitive. Hence it cannot be a equivalence relation.

Problem 2

Let A be a ring, $f \in A$, and M an A-module. Show that the localization $M[1/f]$ is isomorphic to the direct limit

$$L = \lim_{\rightarrow} \left(M \xrightarrow{f} M \xrightarrow{f} M \xrightarrow{f} \cdots \right).$$

Proof: Let $\{M_i\}_{i=1}^{\infty}$ be a collection of copies of M and for $x_i \in M_i$, x_i is a copy of $x \in M$ that is in M_i.

$$L = \bigsqcup_{i=1}^{\infty} M_i / \left(x_i \sim y_j \Leftrightarrow \exists k \geq i, j \ni f^{k-i} x_i = f^{k-j} y_j \right)$$

Let

$$T : L \to M\left[\frac{1}{f}\right]$$
$$x_i \to \frac{x_i}{f^i}$$

Well defined: Suppose $x_i \sim y_j$, thus exists $k \geq i, j$ such that

$$f^{k-i} x_i = f^{k-j} y_j.$$

Furthermore, note that $x_i \sim f^{k-i} x_i = f^{k-j} y_j \sim y_j$ and that $f^{k-i} x_i = f^{k-j} y_j \in M_k$.

$$T\left(f^{k-i} x_i\right) = \frac{f^{k-i} x_i}{f^k} = \frac{x_i}{f^i} = T(x_i)$$

Thus $T\big(f^{k-i}x_i\big) = T(x_i)$. By a symmetric argument, $T\big(f^{k-j}y_j\big) = T\big(y_j\big)$. Thus

$$T(x_i) = T\big(f^{k-i}x_i\big) = T\big(f^{k-j}y_j\big) = T\big(y_j\big).$$

Thus T is well defined.

A linear: Let $k \geq i, j$ thus

$$T\big(x_i + y_j\big) = \frac{f^{k-i}x_i + f^{k-j}y_j}{f^k} = \frac{x}{f^i} + \frac{y}{f^j} = T(x_i) + T\big(y_j\big)$$

and

$$T(ax_i) = \frac{ax}{f^i} = aT(x_i)$$

Injective: Suppose $T(x_i) = 0$. Then

$$0 = \frac{x}{f^i} \Rightarrow \frac{0}{f} = \frac{x}{f^i}.$$

Thus there exists a $n \in \mathbb{N}$ such that $f^n\big(fx - f^i * 0\big) = 0 \Rightarrow f^{n+1}x = 0 \Rightarrow x_i = 0 \in L$.

Surjective: $\forall r \in M\left[\frac{1}{f}\right]$,

$$r = \frac{m}{f^k}$$

for $m \in M$ and $k \in \mathbb{N}_0$. $T(m_k) = \frac{m}{f^k}$.

Thus the map is a isomorphism as a $A\left[\frac{1}{f}\right]$ module. □

Problem 3

Let R be a ring, $f \in R$. Show that the localization $R[1/f]$ is isomorphic to $R[x]/(fx-1)$ (as an R-algebra).

Proof: Let $\varphi : R \to R[x]/(fx-1)$ where $r \to r$. φ is a ring homomorphism. $\varphi(f) = f \in R[x]/(fx-1)$. Note that

$$fx = 1 \Rightarrow f \text{ is unit in } R[x]/(fx-1).$$

By the universal property, $\exists \tilde{\varphi} : R\left[\frac{1}{f}\right] \to R[x]/(fx-1)$ a ring homomorphism, note under this map $\frac{1}{f} \to x$. We also want to check that its a R-algebra map, this $\tilde{\varphi}\big(r\frac{q}{f^k}\big) = \tilde{\varphi}\big(\frac{rq}{f^k}\big) = rqx^k = r\tilde{\varphi}\big(\frac{q}{f^k}\big)$.

By the universal property of the polynomial ring, there exists a unique R-algebra homomorphism:

$$\tilde{\psi} : R[x] \to R\left[\frac{1}{f}\right]$$
$$r \to \frac{r}{1}$$
$$x \to \frac{1}{f}.$$

$$\begin{aligned}\tilde{\psi}(fx-1) &= \tilde{\psi}(f)\tilde{\psi}(x) - 1 \\ &= f * \frac{1}{f} - 1 = 0 \Rightarrow (fx-1) \subseteq \ker\tilde{\psi} \Rightarrow \tilde{\psi} \text{ factors through } \tilde{\psi} : \frac{R[x]}{(fx-1)} \to R\left[\frac{1}{f}\right]\end{aligned}$$

Note that

$$\begin{aligned}(\varphi \circ \psi)\left(\frac{q}{f^k}\right) &= \varphi\big(qx^k\big) \\ &= q * \frac{1}{f^k} \\ &= \frac{q}{f^k}\end{aligned}$$

and

$$\begin{aligned}(\varphi \circ \psi)\left(\sum_{k=0}^{n} a_k x^k\right) &= \varphi\left(\sum_{k=1}^{n} \frac{a_k}{x^k}\right) \\ &= \sum_{k=1}^{n} a_k x^k\end{aligned}$$

□

Problem 4

Let R be a ring, $S \subset R$ a multiplicatively closed subset, M an R-module. Suppose that for every $s \in S$, multiplication by s is a bijection from M to itself. Show that the natural map $M \to M\big[S^{-1}\big]$ is an isomorphism. As a sample application, show that the localization of the $\mathbb{Z}$-module $\mathbb{Z}/8$ at the prime ideal $(2) \subset \mathbb{Z}$ is simply $\mathbb{Z}/8$.

Proof: Let $\varphi : M \to M\big[S^{-1}\big]$ where $m \to \frac{m}{1}$.

Injective: Suppose

$$\varphi(m) = 0,$$

thus exists a $s \in S$ such that

$$sm = 0.$$

. Since $s \times \cdot : M \to M$ is bjective hence injective, $m = 0$. Thus $\ker\varphi = \{0\}$.

Surjective: $\forall \frac{a}{s} \in M\big[S^{-1}\big]$, $\exists! b \in M$ such that $sb = a$. Thus

$$\varphi(b) = \frac{b}{1} = \frac{a}{s}.$$

Hence φ is a isomorphism.

Localizing at (2) in $\mathbb{Z}/8$: $S = \mathbb{Z} - (2) = \{1, 3, 5, ..., 2n+1, ...\}$. All odd number are units in $\mathbb{Z}/8$, thus multiplication by them simply permutes the elements of $\mathbb{Z}/8$. Hence localizing $\mathbb{Z}/8$ at two doesn't do anything ie

$$\left(\frac{\mathbb{Z}}{8}\right)[\mathbb{Z} - (2)] = \frac{\mathbb{Z}}{8}$$

□

Problem 5

A *finite algebra* over a ring A means an A-algebra which is finitely generated as an A-module. Show that a finite flat $\mathbb{Z}$-algebra is free as a $\mathbb{Z}$-module, and give an example of a flat $\mathbb{Z}$-algebra which is not free as a $\mathbb{Z}$-module. Is a flat $\mathbb{Z}$-algebra of finite type always free as a $\mathbb{Z}$-module?

Proof: Let A be a flat finite $\mathbb{Z}$ algebra. By the structure theorem for finitely generated $\mathbb{Z}$ modules, $A = \mathbb{Z}^d \oplus T(A)$ where $d = \text{rank}_{\mathbb{Z}}(A)$ and $T(A)$ is the torsion of A where $T(A) = \bigoplus_{i=1}^{k} \mathbb{Z}/p^{r_i}$ for $k \geq 0$. Assume the contrary $k > 0$. Consider the short exact sequence

$$0 \to \mathbb{Z}^k \xrightarrow{D} \mathbb{Z}^k \to T(A) \to 0.$$

where $D = \text{diag}\big(p_1^{r_1}, ..., p_k^{r_k}\big)$.

Apply the $\otimes_{\mathbb{Z}} A$ functor, we should have

$$0 \to A^k \xrightarrow{D \otimes 1} A^k \to T(A)^d \oplus T(A)^{\otimes 2} \to 0$$

Now take $x \in A$, a generator for the $\mathbb{Z}/p_i^{r_i}$ summand. Thetn $x \neq 0$, but $p_i^{r_i} x = 0$. So $(x, 0, ..., 0) \in \ker(D \otimes 1)$, contradicting injectivity, meaning $k = 0$ and $T(A) = 0$ and $A = \mathbb{Z}^d$.

$\mathbb{Q}$ is a flat $\mathbb{Z}$ algebra, but isn't free as a $\mathbb{Z}$-module.

$$\mathbb{Z}\left[\frac{1}{2}\right] \cong \frac{\mathbb{Z}[x]}{(2x-1)}$$

and thus $\mathbb{Z}\left[\frac{1}{2}\right]$ is a $\mathbb{Z}$-algebra of finite type but isn't free. $\mathbb{Z}\left[\frac{1}{2}\right]$ is flat and hence is a counterexample to the claim that all finite type $\mathbb{Z}$-algebras are always free. □